



Compounds of Transitional elements

61. Which of the following imparts green colour to the glass
 (a) Cu_2O (b) CdS
 (c) MnO_2 (d) Cr_2O_3
62. On the heating copper nitrate strongly, is finally obtained
 (a) Copper
 (b) Copper oxide
 (c) Copper nitrate
 (d) Copper nitride
63. On adding KI to a solution of copper sulphate
 (a) Cupric oxide is precipitated
 (b) Metallic copper is precipitated
 (c) Cuprous iodide is precipitated with liberation of iodine
 (d) No change occurs
64. Which of the following statements is correct about equivalent weight of $KMnO_4$
 (a) It is one third of its molecular weight in alkaline medium
 (b) It is one fifth of its molecular weight in alkaline medium
 (c) It is equal to its molecular weight in acidic medium
 (d) It is one third of its molecular weight in acidic medium
65. The reaction of $K_2Cr_2O_7$ with $NaCl$ and conc. H_2SO_4 gives
 (a) $CrCl_3$ (b) $CrOCl_2$
 (c) CrO_2Cl_2 (d) Cr_2O_3
66. Silver nitrate is supplied in coloured bottles because it is
 (a) Oxidised in air
 (b) Decomposes in sunlight
 (c) Explosive in sunlight
 (d) Reactive towards air in sunlight
67. A nitrate when mixed with common salt gives a white precipitate which is soluble in dilute NH_4OH . It is the nitrate of
 (a) Copper (b) Mercury
 (c) Silver (d) Gold
68. Which one of the following is **lunar caustic**
 (a) $AgNO_3$ (b) Cu_2Cl_2
 (c) $CuCl_2$ (d) Hg_2Cl_2
69. Invar, an alloy of Fe and Ni is used in watches and meter scale, its characteristic property is
 (a) Small coefficient of expansion
 (b) Resistance to corrosion
 (c) Hardness and elasticity
 (d) Magnetic nature
70. The extraction of nickel involves
 (a) The formation of $Ni(CO)_4$
 (b) The decomposition of $Ni(CO)_4$
 (c) The formation and thermal decomposition of $Ni(CO)_4$
 (d) The formation and catalytic decomposition of $Ni(CO)_4$



71. On adding excess of NH_3 solution to $CuSO_4$ solution, the dark blue colour is due to
 (a) $[Cu(NH_3)_4]^{++}$
 (b) $[Cu(NH_3)_2]^{++}$
 (c) $[Cu(NH_3)]^+$
 (d) None of the above
72. If M is the molecular weight of $KMnO_4$, its equivalent weight will be when it is converted into K_2MnO_4
 (a) M (b) $M/3$
 (c) $M/5$ (d) $M/7$
73. While writing the formula of ferrous oxide it is written as (FeO) , because it is
 (a) Non-stoichiometric
 (b) Non-existent
 (c) Paramagnetic
 (d) Ferromagnetic
74. Which of the following exhibit maximum oxidation state of vanadium
 (a) $VOCl_3$ (b) VCl_4
 (c) VCl_3 (d) VCl_2
75. **Prussian blue** is due to the formation of
 (a) $Fe_4[Fe(CN)_6]_3$
 (b) $Fe_2[Fe(CN)_6]$
 (c) $Fe_3[Fe(CN)_6]$
 (d) $Fe[Fe(CN)_6]_3$
76. The Nessler's reagent contains
 (a) Hg_2^{++} (b) Hg^{++}
 (c) HgI_2^{--} (d) HgI_4^{--}
77. Formula of ferric sulphate is
 (a) $FeSO_4$ (b) $Fe(SO_4)_2$
 (c) Fe_2SO_4 (d) $Fe_2(SO_4)_3$
78. When $CuSO_4$ is hydrated, then it becomes
 (a) Acidic
 (b) basic
 (c) Neutral
 (d) Amphoteric
79. Silvering of mirror is done by
 (a) $AgNO_3$ (b) Ag_2O_3
 (c) Fe_2O_3 (d) Al_2O_3
80. The colour of $K_2Cr_2O_7$ changes from red orange to lemon yellow on treatment with aqueous KOH because of
 (a) The reduction of Cr^{VI} to Cr^{III}
 (b) The formation of chromium hydroxide
 (c) The conversion of dichromate to chromate
 (d) The oxidation of potassium hydroxide to potassium peroxide

